

General Physics II — Homework #1

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September 18, 2026

Problem 1.

Consider the scalar field

$$f(x, y, z) = x^2 y e^z + y z^3$$

and the vector field

$$\vec{A}(x, y, z) = (xy + z^2) \hat{x} + (x^2 - yz) \hat{y} + (2xz + y^2) \hat{z}.$$

Calculate the following quantities:

- 1) the gradient ∇f
- 2) the Laplacian $\nabla^2 f$
- 3) the divergence $\nabla \cdot \vec{A}$
- 4) the curl $\nabla \times \vec{A}$

Then explicitly verify that

$$\nabla \times (\nabla f) = 0 \quad \text{and} \quad \nabla \cdot (\nabla \times \vec{A}) = 0.$$

Solution.

1)

$$\nabla f = \frac{\partial f}{\partial x} \hat{x} + \frac{\partial f}{\partial y} \hat{y} + \frac{\partial f}{\partial z} \hat{z} \tag{1}$$

$$= 2xye^z \hat{x} + (x^2 e^z + z^3) \hat{y} + (x^2 y e^z + 3yz^2) \hat{z} \tag{2}$$

2)

$$\nabla^2 f = \nabla \cdot (\nabla f) = \nabla \cdot (2xye^z \hat{x} + (x^2 e^z + z^3) \hat{y} + (x^2 y e^z + 3yz^2) \hat{z}) \tag{3}$$

$$= \frac{\partial}{\partial x} 2xye^z + \frac{\partial}{\partial y} (x^2 e^z + z^3) + \frac{\partial}{\partial z} (x^2 y e^z + 3yz^2) \tag{4}$$

$$= 2ye^z + 0 + (x^2 y e^z + 6yz) \tag{5}$$

$$= (x^2 + 2)ye^z + 6yz \tag{6}$$

3)

$$\nabla \cdot \vec{A} = \frac{\partial}{\partial x} (xy + z^2) + \frac{\partial}{\partial y} (x^2 - yz) + \frac{\partial}{\partial z} (2xz + y^2) \tag{7}$$

$$= y - z + 2x \tag{8}$$

$$= 2x + y - z \tag{9}$$

4)

$$\nabla \times \vec{A} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{x} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \hat{y} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{z} \quad (10)$$

$$= (2y + y)\hat{x} + (2z - 2z)\hat{y} + (2x - x)\hat{z} \quad (11)$$

$$= 3y\hat{x} + x\hat{z} \quad (12)$$

Verification:

$$\nabla \times (\nabla f) = \nabla \times (2xye^z\hat{x} + (x^2e^z + z^3)\hat{y} + (x^2ye^z + 3yz^2)\hat{z}) \quad (13)$$

$$\left(\frac{\partial}{\partial y}(x^2ye^z + 3yz^2) - \frac{\partial}{\partial z}(x^2e^z + z^3) \right) \hat{x} +$$

$$= \left(\frac{\partial}{\partial z}2xye^z - \frac{\partial}{\partial x}(x^2ye^z + 3yz^2) \right) \hat{y} + \quad (14)$$

$$\left(\frac{\partial}{\partial x}(x^2e^z + z^3) - \frac{\partial}{\partial y}2xye^z \right) \hat{z}$$

$$= ((x^2e^z + 3z^2) - (x^2e^z + 3z^2))\hat{x} + (2xye^z - 2xye^z)\hat{y} + (2xe^z - 2xe^z)\hat{z} \quad (15)$$

$$= \vec{0} \quad (16)$$

$$\nabla \cdot (\nabla \times \vec{A}) = \nabla \cdot (3y\hat{x} + x\hat{z}) \quad (17)$$

$$= \frac{\partial}{\partial x}(3y) + \frac{\partial}{\partial z}(x) \quad (18)$$

$$= 0 \quad (19)$$

Answer: $\nabla f = 2xye^z\hat{x} + (x^2e^z + z^3)\hat{y} + (x^2ye^z + 3yz^2)\hat{z}$, $\nabla^2 f = (x^2 + 2)ye^z + 6yz$, $\nabla \cdot \vec{A} = 2x + y - z$, $\nabla \times \vec{A} = 3y\hat{x} + x\hat{z}$

Problem 2.

Let ϕ and ψ be smooth scalar fields, and let $\vec{A}$ and $\vec{B}$ be smooth vector fields. Verify the following vector identities:

$$1) \nabla^2(\phi\psi) = \phi\nabla^2\psi + \psi\nabla^2\phi + 2\nabla\phi \cdot \nabla\psi.$$

$$2) \nabla \cdot (\phi\nabla\psi - \psi\nabla\phi) = \phi\nabla^2\psi - \psi\nabla^2\phi.$$

$$3) \nabla(\vec{A} \cdot \vec{B}) = (\vec{A} \cdot \nabla)\vec{B} + (\vec{B} \cdot \nabla)\vec{A} + \vec{A} \times (\nabla \times \vec{B}) + \vec{B} \times (\nabla \times \vec{A}).$$

For part (3), verify the identity explicitly using Cartesian components.

Solution.

1)

$$\nabla^2(\phi\psi) = \frac{\partial^2(\phi\psi)}{\partial x^2} + \frac{\partial^2(\phi\psi)}{\partial y^2} + \frac{\partial^2(\phi\psi)}{\partial z^2} \quad (20)$$

$$= \sum_{i=x,y,z} \left(\psi \frac{\partial^2 \phi}{\partial i^2} + 2 \frac{\partial \phi}{\partial i} \frac{\partial \psi}{\partial i} + \phi \frac{\partial^2 \psi}{\partial i^2} \right) \quad (21)$$

$$= \sum_{i=x,y,z} \psi \frac{\partial^2 \phi}{\partial i^2} + 2 \sum_{i=x,y,z} \frac{\partial \phi}{\partial i} \frac{\partial \psi}{\partial i} + \sum_{i=x,y,z} \phi \frac{\partial^2 \psi}{\partial i^2} \quad (22)$$

$$= \psi \sum_{i=x,y,z} \frac{\partial^2 \phi}{\partial i^2} + \phi \sum_{i=x,y,z} \frac{\partial^2 \psi}{\partial i^2} + 2 \left(\sum_{i=x,y,z} \frac{\partial \phi}{\partial i} \hat{i} \right) \cdot \left(\sum_{i=x,y,z} \frac{\partial \psi}{\partial i} \hat{i} \right) \quad (23)$$

$$= \psi \nabla^2 \phi + \phi \nabla^2 \psi + 2 \nabla \phi \cdot \nabla \psi \quad (24)$$

2)

$$\nabla \cdot (\phi \nabla \psi - \psi \nabla \phi) = \nabla \cdot \left(\sum_{i=x,y,z} \left(\phi \frac{\partial \psi}{\partial i} - \psi \frac{\partial \phi}{\partial i} \right) \hat{i} \right) \quad (25)$$

$$= \sum_{i=x,y,z} \frac{\partial}{\partial i} \left(\phi \frac{\partial \psi}{\partial i} - \psi \frac{\partial \phi}{\partial i} \right) \quad (26)$$

$$= \sum_{i=x,y,z} \left(\left(\frac{\partial \phi}{\partial i} \frac{\partial \psi}{\partial i} + \phi \frac{\partial^2 \psi}{\partial i^2} \right) - \left(\frac{\partial \psi}{\partial i} \frac{\partial \phi}{\partial i} + \psi \frac{\partial^2 \phi}{\partial i^2} \right) \right) \quad (27)$$

$$= \sum_{i=x,y,z} \phi \frac{\partial^2 \psi}{\partial i^2} - \sum_{i=x,y,z} \psi \frac{\partial^2 \phi}{\partial i^2} \quad (28)$$

$$= \phi \nabla^2 \psi - \psi \nabla^2 \phi \quad (29)$$

3)

$$(A \cdot \nabla)B = \sum_{i=x,y,z} \left(A_x \frac{\partial B_i}{\partial x} + A_y \frac{\partial B_i}{\partial y} + A_z \frac{\partial B_i}{\partial z} \right) \hat{i} \quad (30)$$

$$(B \cdot \nabla)A = \sum_{i=x,y,z} \left(B_x \frac{\partial A_i}{\partial x} + B_y \frac{\partial A_i}{\partial y} + B_z \frac{\partial A_i}{\partial z} \right) \hat{i} \quad (31)$$

$$A \times (\nabla \times B) = A \times \left(\left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) \hat{x} + \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) \hat{y} + \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right) \hat{z} \right) \quad (32)$$

$$= \left(A_y \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right) - A_z \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) \right) \hat{x} + \quad (33)$$

$$\left(A_z \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) - A_x \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right) \right) \hat{y} +$$

$$\left(A_x \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) - A_y \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) \right) \hat{z}$$

$$B \times (\nabla \times A) = B \times \left(\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{x} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \hat{y} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{z} \right) \quad (34)$$

$$= \left(B_y \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) - B_z \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \right) \hat{x} + \quad (35)$$

$$\left(B_z \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) - B_x \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \right) \hat{y} +$$

$$\left(B_x \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) - B_y \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \right) \hat{z}$$

First, we verify the x component

$$x \text{ component of R.H.S} = \left(A_x \frac{\partial B_x}{\partial x} + A_y \frac{\partial B_x}{\partial y} + A_z \frac{\partial B_x}{\partial z} \right) + \quad (36)$$

$$\begin{aligned} & \left(B_x \frac{\partial A_x}{\partial x} + B_y \frac{\partial A_x}{\partial y} + B_z \frac{\partial A_x}{\partial z} \right) + \\ & \left(A_y \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right) - A_z \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) \right) + \\ & \left(B_y \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) - B_z \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \right) \\ & = A_x \frac{\partial B_x}{\partial x} + A_y \frac{\partial B_x}{\partial y} + A_z \frac{\partial B_x}{\partial z} + \quad (37) \\ & B_x \frac{\partial A_x}{\partial x} + B_y \frac{\partial A_x}{\partial y} + B_z \frac{\partial A_x}{\partial z} + \\ & A_y \frac{\partial B_y}{\partial x} - A_y \frac{\partial B_x}{\partial y} - A_z \frac{\partial B_x}{\partial z} + A_z \frac{\partial B_z}{\partial x} + \\ & B_y \frac{\partial A_y}{\partial x} - B_y \frac{\partial A_x}{\partial y} - B_z \frac{\partial A_x}{\partial z} + B_z \frac{\partial A_z}{\partial x} \end{aligned}$$

$$\begin{aligned} & = A_y \frac{\partial B_y}{\partial x} + A_z \frac{\partial B_z}{\partial x} + B_y \frac{\partial A_y}{\partial x} + B_z \frac{\partial A_z}{\partial x} + A_x \frac{\partial B_x}{\partial x} + B_x \frac{\partial A_x}{\partial x} \quad (38) \\ & = \left(A_x \frac{\partial B_x}{\partial x} + B_x \frac{\partial A_x}{\partial x} \right) + \left(A_y \frac{\partial B_y}{\partial x} + B_y \frac{\partial A_y}{\partial x} \right) + \left(A_z \frac{\partial B_z}{\partial x} + B_z \frac{\partial A_z}{\partial x} \right) \quad (39) \end{aligned}$$

$$= \frac{\partial}{\partial x} (A_x B_x) + \frac{\partial}{\partial x} (A_y B_y) + \frac{\partial}{\partial x} (A_z B_z) \quad (40)$$

$$= \frac{\partial}{\partial x} (\vec{A} \cdot \vec{B}) \quad (41)$$

$$= (\nabla(\vec{A} \cdot \vec{B}))_x \quad (42)$$

y and z components follow by cyclic permutation.

Answer: All three identities verified.

Problem 3.

Consider the vector field

$$\vec{F} = x^2 \hat{x} + y^2 \hat{y} + z^2 \hat{z}.$$

Let V be the tetrahedral region in the first octant bounded by $x \geq 0$, $y \geq 0$, $z \geq 0$, $x + y + z \leq a$, where $a > 0$. Check the divergence theorem

$$\iiint_V (\nabla \cdot \vec{F}) dV = \iint_{\partial V} \vec{F} \cdot d\vec{S}$$

by evaluating the two sides independently.

1) Calculate $\iiint_V (\nabla \cdot \vec{F}) dV$.

2) Calculate the flux of $\vec{F}$ through each of the four faces of the tetrahedron.

3) Show explicitly that the total outward flux agrees with the result in part (1).

Hint. On the inclined surface $x + y + z = a$, write $z = a - x - y$. For the outward orientation, $d\vec{S} = (\hat{x} + \hat{y} + \hat{z}) dx dy$.

Solution.

1)

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(y^2) + \frac{\partial}{\partial z}(z^2) \quad (43)$$

$$= 2(x + y + z) \quad (44)$$

$$\iiint_V (\nabla \cdot \vec{F}) dV = 2 \iiint_V (x + y + z) dV \quad (45)$$

$$(46)$$

By symmetry,

$$\iiint_V x dV = \iiint_V y dV = \iiint_V z dV \quad (47)$$

$$\iiint_V x dV = \int_0^a dx \int_0^{a-x} dy \int_0^{a-x-y} dz x \quad (48)$$

$$= \int_0^a dx \int_0^{a-x} dy x(a - x - y) \quad (49)$$

$$= \int_0^a dx \frac{1}{2} x(a - x)^2 \quad (50)$$

$$= \frac{a^4}{24} \quad (51)$$

$$2 \iiint_V (x + y + z) dV = 2 \cdot 3 \iiint_V x dV \quad (52)$$

$$= \frac{a^4}{4} \quad (53)$$

2) On the xy surface,

$$\Phi_{xy} = \int_0^a dx \int_0^{a-x} dy \vec{F} \cdot (-\hat{z}) \quad (54)$$

$$= \int_0^a dx \int_0^{a-x} dy -z^2 \quad (55)$$

$$= \int_0^a dx \int_0^{a-x} dy 0 \quad (56)$$

$$= 0 \quad (57)$$

By symmetry, $\Phi_{yz} = 0$, $\Phi_{xz} = 0$, but on the inclined surface,

$$\Phi_I = \int_0^a dx \int_0^{a-x} dy \vec{F} \cdot (\hat{x} + \hat{y} + \hat{z}) \quad (58)$$

$$= \int_0^a dx \int_0^{a-x} dy (x^2 + y^2 + (a - x - y)^2) \quad (59)$$

$$= \int_0^a dx x^2(a - x) + \frac{2}{3}(a - x)^3 \quad (60)$$

$$= \frac{a^4}{4} \quad (61)$$

3)

$$\iint_{\partial V} \vec{F} \cdot d\vec{S} = 0 + 0 + 0 + \frac{a^4}{4} \quad (62)$$

$$= \frac{a^4}{4} \quad (63)$$

$$= \iiint_V (\nabla \cdot \vec{F}) dV \quad (64)$$

Answer: Both sides equal $\frac{a^4}{4}$.

Problem 4.

Consider the vector field

$$\vec{A} = -y \hat{x} + x \hat{y} + (x^2 + y^2) \hat{z}.$$

Let S be the paraboloidal surface $z = 1 - x^2 - y^2$, $x^2 + y^2 \leq 1$, oriented upward. The boundary C of S is the circle $x^2 + y^2 = 1$, $z = 0$, oriented counterclockwise when viewed from the positive z -axis. Check Stokes' theorem

$$\oint_C \vec{A} \cdot d\vec{l} = \iint_S (\nabla \times \vec{A}) \cdot d\vec{S}$$

by calculating the two sides independently.

1) Parameterize the boundary C and calculate $\oint_C \vec{A} \cdot d\vec{l}$.

2) Calculate $\nabla \times \vec{A}$.

3) Calculate $\iint_S (\nabla \times \vec{A}) \cdot d\vec{S}$ and verify Stokes' theorem.

Hint. For a surface written as $z = z(x, y)$ with upward orientation,

$$d\vec{S} = \left(-\frac{\partial z}{\partial x} \hat{x} - \frac{\partial z}{\partial y} \hat{y} + \hat{z} \right) dx dy.$$

Solution.

1) Let $x = \cos \theta$ and $y = \sin \theta$, then

$$\vec{A} = -\sin \theta \hat{x} + \cos \theta \hat{y} + \hat{z} \quad (65)$$

on boundary, and $d\vec{l} = \hat{\theta} d\theta = (-\sin \theta \hat{x} + \cos \theta \hat{y}) d\theta$, hence

$$\oint_C \vec{A} \cdot d\vec{l} = \int_0^{2\pi} \sin^2 \theta + \cos^2 \theta + 0 \quad (66)$$

$$= 2\pi \quad (67)$$

2)

$$\nabla \times \vec{A} = \left(\frac{\partial(x^2 + y^2)}{\partial y} - \frac{\partial x}{\partial z} \right) \hat{x} + \left(\frac{\partial(-y)}{\partial z} - \frac{\partial(x^2 + y^2)}{\partial x} \right) \hat{y} + \left(\frac{\partial x}{\partial x} - \frac{\partial(-y)}{\partial y} \right) \hat{z} \quad (68)$$

$$= 2y\hat{x} - 2x\hat{y} + 2\hat{z} \quad (69)$$

$$(70)$$

3)

$$d\vec{S} = \left(-\frac{\partial z}{\partial x} \hat{x} - \frac{\partial z}{\partial y} \hat{y} + \hat{z} \right) dx dy \quad (71)$$

$$= (2x\hat{x} + 2y\hat{y} + \hat{z}) dx dy \quad (72)$$

Therefore

$$\iint_S (\nabla \times \vec{A}) \cdot d\vec{S} = \iint_{x^2+y^2 \leq 1} (4xy - 4xy + 2) dx dy \quad (73)$$

$$= 2 \iint_{x^2+y^2 \leq 1} dx dy \quad (74)$$

$$= 2\pi \quad (75)$$

$$= \oint_C \vec{A} \cdot d\vec{l} \quad (76)$$

Answer: Both sides equal 2π .

Problem 5.

The cylindrical coordinates (r, ϕ, z) are related to Cartesian coordinates by

$$x = r \cos \phi, \quad y = r \sin \phi, \quad z = z.$$

Starting from the Cartesian coordinate system, derive the following results.

1) Show that

$$\frac{\partial}{\partial x} = \cos \phi \frac{\partial}{\partial r} - \frac{\sin \phi}{r} \frac{\partial}{\partial \phi}, \quad \frac{\partial}{\partial y} = \sin \phi \frac{\partial}{\partial r} + \frac{\cos \phi}{r} \frac{\partial}{\partial \phi}.$$

2) Derive the gradient operator in cylindrical coordinates:

$$\nabla = \hat{e}_r \frac{\partial}{\partial r} + \hat{e}_\phi \frac{1}{r} \frac{\partial}{\partial \phi} + \hat{e}_z \frac{\partial}{\partial z}.$$

3) Derive the Laplacian operator:

$$\nabla^2 = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2}{\partial \phi^2} + \frac{\partial^2}{\partial z^2}.$$

4) Using the result above, show that $f_n(r, \phi) = r^n \cos(n\phi)$ satisfies $\nabla^2 f_n = 0$ for $r > 0$ and any positive integer n .

Solution.

1) First, we have

$$r = \sqrt{x^2 + y^2} \quad (77)$$

$$\phi = \arctan\left(\frac{y}{x}\right) \quad (78)$$

then

$$\frac{\partial}{\partial x} = \frac{\partial r}{\partial x} \frac{\partial}{\partial r} + \frac{\partial \phi}{\partial x} \frac{\partial}{\partial \phi} \quad (79)$$

$$= \frac{x}{\sqrt{x^2 + y^2}} \frac{\partial}{\partial r} - \frac{y}{x^2 + y^2} \frac{\partial}{\partial \phi} \quad (80)$$

$$= \frac{r \cos \phi}{r} \frac{\partial}{\partial r} - \frac{r \sin \phi}{r^2} \frac{\partial}{\partial \phi} \quad (81)$$

$$= \cos \phi \frac{\partial}{\partial r} - \frac{\sin \phi}{r} \frac{\partial}{\partial \phi} \quad (82)$$

$$\frac{\partial}{\partial y} = \frac{\partial r}{\partial y} \frac{\partial}{\partial r} + \frac{\partial \phi}{\partial y} \frac{\partial}{\partial \phi} \quad (83)$$

$$= \frac{y}{\sqrt{x^2 + y^2}} \frac{\partial}{\partial r} + \frac{x}{x^2 + y^2} \frac{\partial}{\partial \phi} \quad (84)$$

$$= \frac{r \sin \phi}{r} \frac{\partial}{\partial r} + \frac{r \cos \phi}{r^2} \frac{\partial}{\partial \phi} \quad (85)$$

$$= \sin \phi \frac{\partial}{\partial r} + \frac{\cos \phi}{r} \frac{\partial}{\partial \phi} \quad (86)$$

2)

$$\begin{pmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} = \begin{pmatrix} \cos \phi \frac{\partial}{\partial x} + \sin \phi \frac{\partial}{\partial y} \\ -\sin \phi \frac{\partial}{\partial x} + \cos \phi \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \quad (87)$$

$$= \begin{pmatrix} \cos^2 \phi \frac{\partial}{\partial r} - \frac{\sin \phi \cos \phi}{r} \frac{\partial}{\partial \phi} + \sin^2 \phi \frac{\partial}{\partial r} + \frac{\sin \phi \cos \phi}{r} \frac{\partial}{\partial \phi} \\ -\sin \phi \cos \phi \frac{\partial}{\partial r} + \frac{\sin^2 \phi}{r} \frac{\partial}{\partial \phi} + \sin \phi \cos \phi \frac{\partial}{\partial r} + \frac{\cos^2 \phi}{r} \frac{\partial}{\partial \phi} \\ \frac{\partial}{\partial z} \end{pmatrix} \quad (88)$$

$$= \begin{pmatrix} \frac{\partial}{\partial r} \\ \frac{1}{r} \frac{\partial}{\partial \phi} \\ \frac{\partial}{\partial z} \end{pmatrix} \quad (89)$$

$$= \hat{e}_r \frac{\partial}{\partial r} + \hat{e}_\phi \frac{1}{r} \frac{\partial}{\partial \phi} + \hat{e}_z \frac{\partial}{\partial z} \quad (90)$$

3) First, we have

$$\frac{\partial}{\partial \phi} \hat{e}_r = \hat{e}_\phi \quad (91)$$

$$\frac{\partial}{\partial \phi} \hat{e}_\phi = -\hat{e}_r \quad (92)$$

thus

$$\nabla^2 = \nabla \cdot \nabla = \left(\hat{e}_r \frac{\partial}{\partial r} + \hat{e}_\phi \frac{1}{r} \frac{\partial}{\partial \phi} + \hat{e}_z \frac{\partial}{\partial z} \right) \cdot \left(\hat{e}_r \frac{\partial}{\partial r} + \hat{e}_\phi \frac{1}{r} \frac{\partial}{\partial \phi} + \hat{e}_z \frac{\partial}{\partial z} \right) \quad (93)$$

$$= \hat{e}_r \frac{\partial}{\partial r} \left(\hat{e}_r \frac{\partial}{\partial r} \right) + \hat{e}_\phi \frac{1}{r} \frac{\partial}{\partial \phi} \left(\hat{e}_r \frac{\partial}{\partial r} \right) + \hat{e}_\phi \frac{1}{r} \frac{\partial}{\partial \phi} \left(\hat{e}_\phi \frac{1}{r} \frac{\partial}{\partial \phi} \right) + \hat{e}_z \frac{\partial}{\partial z} \left(\hat{e}_z \frac{\partial}{\partial z} \right) \quad (94)$$

$$= \frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \phi^2} + \frac{\partial^2}{\partial z^2} \quad (95)$$

$$= \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2}{\partial \phi^2} + \frac{\partial^2}{\partial z^2} \quad (96)$$

4)

$$\nabla^2 f_n = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} (r^n \cos(n\phi)) \right) + \frac{1}{r^2} \frac{\partial^2}{\partial \phi^2} (r^n \cos(n\phi)) \quad (97)$$

$$= \frac{1}{r} \frac{\partial}{\partial r} (nr^n \cos(n\phi)) + \frac{1}{r^2} \frac{\partial}{\partial \phi} (-nr^n \sin(n\phi)) \quad (98)$$

$$= n^2 r^{n-2} \cos(n\phi) - n^2 r^{n-2} \cos(n\phi) \quad (99)$$

$$= 0 \quad (100)$$

Answer: All four results derived; $\nabla^2 f_n = 0$ verified.